SRI VENKATESWARA COLLEGE OF ENGINEERING

AUTONOMOUS)

Department of Computer Science and Engineering(Artificial Intelligence and Machine Learning)

HygionX – Bridging Artificial Intelligence and Explainable Medicine

Course: B.Tech **Year:** IV **Semester:** I **Section:** B

ABSTRACT

Healthcare chatbots have become increasingly popular in providing medical guidance and emotional support. However, most existing systems, such as Babylon Health, Ada, and Wysa, still face major challenges in delivering personalized and trustworthy healthcare information. They often provide generic responses that overlook a user's medical history, emotional context, or language preferences, leading to limited reliability and user trust.

HygionX – Bridging Artificial Intelligence and Explainable Medicine aims to address these gaps by developing an intelligent healthcare chatbot capable of understanding users in a personalized, transparent, and secure manner. The system not only answers queries but also considers a patient's past health data, existing conditions, and symptoms before providing any suggestion. This ensures the responses are both relevant and context-aware.

The project integrates advanced biomedical language models such as **BioBERT**, **ClinicalBERT**, and **PubMedBERT** for accurate medical understanding, supported by **SQL** and **ElasticSearch** databases for managing patient information and retrieving medical knowledge. An **Explainable AI (XAI)** layer using tools like **SHAP** and **LIME** highlights the reasoning behind every response, helping users understand the "why" behind the chatbot's answers.

By incorporating data privacy, encryption, and ethical validation checks, HygionX ensures secure and reliable interaction with users. Through its focus on personalization, transparency, and continuous learning, the project aims to redefine healthcare chatbots into dependable, explainable, and human-centric digital healthcare companions.

Project Team: 5

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